

Machine Learning Approach to Anti-Money Laundering: A Review

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Abstract—For decades, money laundering has been a global issue, posing a severe threat to economies and societies. Despite the fact that billions of dollars in fines from authorities and the dismantling of illegal businesses continue to make headlines, government, regulatory, and financial bodies are all fighting the problem in their own ways. Traditional techniques of detecting and monitoring unlawful activities, while improved to some extent, are insufficient to keep up with the rising pace of criminal activity and the volume of data and transactions that flow around the world. Machine Learning techniques for anti-money laundering implementation have already been offered, however they are still in the early stages of development. Deep Learning approaches have been discovered to work even better with the enormous datasets that have been provided. This paper attempts to highlight and compare some of the traditional and machine learning based techniques that have been used and proposed to curb money laundering.

Index Terms—machine learning (ML), deep learning (DL), anti-money laundering (AML), fraudulent transactions

I. INTRODUCTION

The processing of criminal proceeds in order to conceal their illegal origins is known as money laundering. Once these funds have been successfully "laundered," the criminal can enjoy them without drawing attention to the underlying activity or the individuals involved. For decades, money laundering has been a global phenomenon that has posed a significant danger to economies and society. Government, regulatory, and financial organizations are all fighting it in their own capacities, yet billions of dollars in fines from authorities and the busting of illicit operations continue to make the news [1]. The damage caused by money laundering has put countries like Nigeria on the world's watch-list. Many transactions originating from Nigeria are usually flagged

for confirmation from the various institutions. Laundered finances can also be used in financing terrorism. In the past, financial institutions (FIs) have used traditional or legacy systems to try to flag money laundering activities however, due to fragmented data, inability to harness such data, and high false-positive rates, these legacy rules-based systems lack holistic and integrated views of parties, accounts, and transactions [2]. Effectively combating financial crime and ensuring regulatory compliance in today's fast-paced and more digital environment necessitates smarter, faster, and more dynamic tools and solutions. This has led to the introduction of more sophisticated anti-money laundering tools.

Anti-money laundering (AML) software enables interested parties to detect, examine, and report suspicious conduct that could be indicative of money laundering, terrorism financing, or fraud. The software is at the heart of an organization's AML compliance operations for the majority of stakeholders. Given the volume of transactions that financial institutions execute on a daily basis, achieving compliance with regulatory standards around the world necessitates sophisticated anti-money laundering software that can analyze transactions from several perspectives and in real time. A solution should also have enough granularity to allow an institution to look at a customer, a transaction, or a series of transactions from a variety of perspectives [3]. Traditionally, banks have relied on rule-based anti-money laundering systems to combat financial crime and comply with anti-money laundering regulations. These systems aid in the identification of cash transactions exceeding threshold values, international fund transfers, and suspicious transactions. These systems raise alerts when suspicious transactions are detected based on pre-defined rules. The Compliance Officer investigates these signals and, if positive, prepares and submits a Suspicious

Matter Report (SMR) to the appropriate regulatory body to ensure compliance reporting. The false positive rate of these warnings is reported to be above 98% [4]. This results in huge operational costs and time delays for transactions that could be critical. By realizing the complexity, volume of transactions, and success of artificial intelligence in other fields, few FIs have begun putting in the effort to automate data and time-intensive tasks using Artificial Intelligence/Machine Learning (AI/ML) technologies for detection of suspicious transactions, to improve operational efficiency and prediction accuracy [5].

This paper aims to highlight and compare some of the traditional and machine learning-based anti-money laundering strategies that have been utilized and suggested. Several articles were examined in order to compare and contrast the existing AML methodologies and how they affect the efficiency and accuracy of the AML technique in question. The goal is to give a structured, thorough assessment of earlier and more recent results in the detection and prevention of money laundering and fraudulent behavior. Related works, existing approaches, merits and demerits of existing AML procedures, and future projections are reviewed in the following sections. Fig. 1 shows how money is laundered and why it gets increasingly difficult for FIs to flag all illicit transactions. Placement involves putting money into the financial system; in layering, money is transferred between separate bank accounts in order to be converted into a different form thereby creating a complex layer that will obscure the source of money and make tracking it difficult; and integration involves transferring money into other forms, such as automobiles and structures. It is near impossible to trace it back to the dirty source after this point.



Fig. 1. Money Laundering process

Using traditional AML systems, Fig. 2 depicts the major features used in tracking and detecting laundered funds. Customer due diligence (CDD) is the process of confirming an intending client's identity by double-checking valid forms of identification; know your customer (KYC) is the practice of verifying a client's identity; customer behavior is the study of a client's transaction patterns; suspicious activity monitoring is the monitoring of a client's account; record-keeping is the keeping of historical accounts and transaction records; case management system is for central data storage and tracking

client data; and watch-list filtering assists FIs in applying required rules and staying current with changes in worldwide regulations and blacklists.

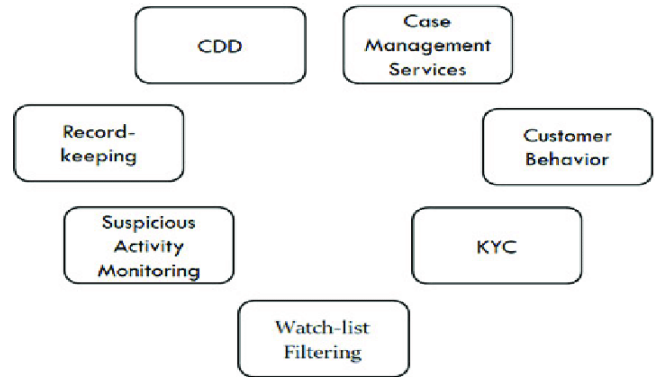


Fig. 2. Anti Money Laundering process

II. LITERATURE REVIEW

Researchers have proposed, worked on and reviewed a number of ways in which machine/deep learning and artificial intelligence can help improve the existing AML systems in place (see Table I).

The **Clustering with sLOPE (CLOPE)** clustering, a fast and scalable algorithm for clustering traditional datasets, is an example of an algorithm for AML typologies. Cao et al. [6] provided a method for recognizing three frequent money laundering instances using the grouping approach-based CLOPE calculation. Money is moved around, money is distributed in small amounts to many beneficiaries, and money is collected from various sources. The experimental results showed that CLOPE could distinguish each suspicious record using only six clusters, without stating which cluster was associated with money laundering. This is regarded as a flaw that renders the method impractical.

A link analysis system that aids in the detection of money laundering was proposed in [7]. Here, a three-component system that aids in the identification of money laundering is presented. The first is clustering; "it entails creating graphs that depict the flow of money and only captures questionable money transactions between groupings of accounts. The second step is to look for common patterns in clusters. Lastly, the third component, also known as data visualization, is concerned with how the result and transaction flow are displayed. The source and destination of a certain banking transaction are required for the clustering graphs to be built, and determining the exact destination can be difficult if the money passes through multiple institutions and countries". This poses a limitation to this approach to AML.

For the behavioral modeling element, the researchers in [8] suggested employing the expectation-maximization (EM) method as the clustering model in discovering fraud. The plan was to use the EM cluster to arrange the data

into a comparable cluster by developing a model based on each bank's past transaction behavior. The author provided no experimental results on the algorithm's performance on synthetic or real-world datasets. This method presupposes that customer transactions follow a normal distribution, which is not true.

With the implementation of the Decision Tree (DT) algorithm, [9] proposed a two-step loan credibility prediction system that made it easy for financial institutions to make the appropriate decision to approve or reject customer loan requests. Credit risk management is crucial for a successful bank loan process, according to the authors. This model is built in five stages: issue comprehension, data comprehension, data filtering, system modeling, and system evaluation.

Under the anomaly detection aspect, [10] presented the suspicious activity reporting using dynamic Bayesian networks (SARDBN). To detect anomalies in transactions, they used a combination of clustering and Bayesian networks. There are three stages to this strategy. To begin, clustering is a method of grouping clients based on the amount of average monthly credit and debit transactions, the frequency of those transactions, and the duration between consecutive transactions. Using the fuzzy c-means method, this phase focused on grouping consumer behavior based on transaction activity. The development of a Dynamic Bayesian network (DBN) on each cluster was determined in the next phase by SARDBN. On three-time slices, each DBN was built using three variables: transaction amount, transaction mode, and transaction period.

The neural network technique was used in [11] to develop an AML service solution for a union bank to detect money laundering on online payments. The suggested method's logical framework has five layers: a database layer, a basic data resource base layer, a data analysis layer, an application service layer, and an interface layer. The database layer collects transaction data. The basic data resource layer then contains a knowledge base, case database, and other useful information that allowed money laundering cases to be discovered. In the data analysis layer, all of the collected data is subsequently converted into meaningful applications. This layer cleans the data before sending it to many agents for analysis, including a neural network agent, an expert system agent, and a data mining agent. The detection component was located at the application service layer, where relevant data from new incoming transactions was extracted and displayed to users in order for them to make a decision.

More intricate and elaborate network structures are designed to better excavate and utilize sequential data. Convolutional Neural Networks (CNNs) are capable of capturing short-term contextual information and can be used to detect financial fraud via convolutional operations. Fu et al. [12] used a CNN-based technique to discover latent fraud tendencies by recombining transaction data to feature matrices, while [13] proposed a meta-learning-based classifier with a feature extraction module and a K-Tuplet Network based on ResNet-34, a common CNN structure.

Wang et al. [14] treated the e-commerce fraud detection problem as a sequence classification problem and used Long Short Term Memory (LSTM) networks to include the users' prior behavior in order to detect fraud on fresh incoming transactions.

Han et al. [15] proposed a Natural Language processing (NLP) based framework that they claimed will "drastically decrease the time and effort of false positive validation (FPV) for AML solutions". They use a distributed architecture to combine different data sources and deep learning-based NLP approaches to produce a recommendation engine to assist human analysts in making more efficient and accurate decisions. "This option looks into using heterogeneous open data (such as social media, news articles, and fraud databases) in AML compliance to make it more robust and up to date; using different levels of sentiment analysis (SA) to identify negative evidence of a target entity; and using name entity recognition (NER) and relation extraction (RE) to build a target entity's relation network and analyze the hidden and complex connections between the target entity and existing suspicious entities".

III. EXISTING METHODOLOGIES

A. Traditional Methods of AML Implementation

Prior to the incorporation of ML in anti-money laundering policies, traditional methods were employed. The most prominent ones include:

- **Rule-based approach:** This is the most traditional AML implementation approach. This approach to AML would flag cash transactions beyond a given currency threshold value, limit transactions to specific countries, select accounts for enhanced monitoring based on customer data, and categorize merchant accounts based on previous transactions. These systems are already widely utilized, but they necessitate a large amount of bank resources to evaluate the transactions that have been detected or banned in order to eliminate false positives. Furthermore, a rules-based AML approach will be unable to react to changes in illegal conduct intended to avoid detection. To put it another way, the rules-based approach is not very flexible [16].
- **Blacklists:** This is quite similar to the rules-based approach. Blacklists are listings of people, countries, or other types of entities that have been blacklisted. Public blacklists, which can be utilized by watch-list filtering systems and are published by governments or economic, political, and law enforcement entities, are one sort of blacklist. Financial institutions can also build private blacklists. Every list has its own structure, although they usually contain some essential information such as the person's complete name, alias, or entity type (Individual, Country, Group, etc.) [17].
- **SWIFT Messages Technology:** The Society for Worldwide Interbank Financial Telecommunication (SWIFT) is

TABLE I
SUMMARY OF REVIEWED PAPERS

Reference	Study	Methodology	Contributions
[6]Cao et al.	Applying data mining in money laundering detection for the Vietnamese banking industry	CLOPE clustering	The CLOPE algorithm was used to implement a system for identifying money laundering in Vietnam's banking industry.
[7]Sepielak et al.	The application of social network analysis algorithms in a system supporting money laundering detection	Clustering graphs, clustering patterns	A link analysis system that aids in the detection of money laundering was proposed.
[8]Chitra et al.	Data mining techniques and its applications in banking sector	SVM, DT, LR	In order to detect fraud, researchers suggested using the expectation-maximization (EM) method as the clustering model.
[9]Chen et al.	Machine learning techniques for anti-money laundering(AML) solutions in suspicious transaction detection: A review	DT, link analysis, behavioural modelling, risk scoring, anomaly detection	Researchers presented a two-step loan credibility prediction method to help financial institutions make informed decisions about whether to approve or reject consumer loan requests.
[10]Raza et al.	Suspicious activity reporting using dynamic Bayesian networks	Bayesian networks, fuzzy c-means clustering	The technique of employing dynamic Bayesian networks to report suspicious activities was demonstrated by the researchers.
[11]Yang et al.	Study on anti-money laundering service system of online payment based on union-bank mode	Neural networks	Researchers built an AML system as a service function and proposed a logical framework of a multi-agent neural network for finding hidden patterns of money laundering.
[12]Fu et al.	Credit card fraud detection using convolutional neural networks	CNNs	To capture the intrinsic patterns of fraud behaviors acquired from labeled data, a CNN-based fraud detection system was proposed by researchers.
[13]Zheng et al.	Federated meta-learning for fraudulent credit card detection	CNNs	Researchers developed a feature extraction module for a meta-learning-based classifier.
[14]Wang et al.	Session-based fraud detection in online e-commerce transactions using recurrent neural networks	RNNs	In order to detect fraud on new incoming transactions, researchers employed LSTM networks that included the users' historical activity.
[15]Han et al.	NextGen AML: Distributed Deep Learning based Language Technologies to Augment Anti Money Investigation	DL, NLP	Researchers developed a recommendation engine to help human analysts make more accurate and efficient conclusions.

a non-profit organization that facilitates secure financial transactions and communications between financial institutions all over the world. Each SWIFT-enabled financial institution has its own code that identifies where the communication originated and for what purpose it was sent. SWIFT messages are pre-formatted communications that can be utilized for a variety of applications. The transaction-ID, sender, and receiver are all important pieces of information [18]. The SWIFT messages are grouped into ten categories allowing the FIs to group the messages under the appropriate categories.

B. Deep Learning Approach to AML Implementation

As shown above, the traditional approaches to AML are severely lacking and resource-intensive. The advent of deep learning techniques in AML implementation has improved the success rate of these detection algorithms.

A number of advancements have been made in AML using various machine/deep learning techniques such as using scalable graph convolutional neural networks (SGCNN), multi-channel convolutional neural networks, autoencoder, graph convolutional neural network and multi-layer perceptron. This is due to the fact that deep learning techniques have been found to give accurate predictions. This paper briefly compares the above-mentioned techniques based on their performance in flagging illicit transactions [19].

SGCNN was created for forensic analysis of data that was vast, dense, and dynamic in nature. A semi-supervised learning model predicts the suspiciousness of a given node using escalated alerts and SMRs as label data. It can handle a big amount of data, but [20] suggests that it can be improved further to save training time. On various data sources such as news, tweets, and social media, the suggested framework for multi-layer CNN conducts multiple levels of sentiment analysis, entity identification, relationship extraction, and link analysis. The system begins by recognizing suspicious transactions, after which important attributes such as name, location, and account details are collected from the flagged transactions. The Stanford NER recognizer and neural relation extraction (NRE) based on the LSTM-CRF framework are used to accomplish Name Entity Recognition (NER). The results were confirmed by AML experts, who stated that compared to the current manual approach to AML inquiry, roughly 30% of the cost can be avoided [15].

The proposed method by [21] has used graph convolutional network (GCN) along with multi-layer perceptron, which has given better results than only GCN as used in the original research paper. The proposed approach is evaluated using elliptic data that belongs to real Bitcoins transactions.

The following table shows a brief comparison between some AML systems:

TABLE II
MERITS AND DEMERITS OF SOME EXISTING AML SYSTEMS

Category	Merits	Demerits
Traditional AML Systems	Adequate ability to detect and flag suspicious activity	Systems not well equipped to recognize and analyze patterns in large volumes of transactions
SGCNN, GNN, KG-based AML Systems	Ability to perform forensic analysis on large, dense and dynamic branched data	Time and space complexity leading to degraded performance which is sensitive in critical systems like AML systems.
Neural relation extraction-based AML Systems	Ability to streamline large volumes of data based on keywords	Natural language processing can be a hindrance to NER-based systems used in transaction monitoring.

IV. RESEARCH GAP

Self-organizing maps (SOMs) are sophisticated data visualization tools that use self-organizing artificial neural networks to minimize the dimensions of data. Though ideally expected to be very effective in understanding high dimensional data, the question of whether SOMs can be utilized in AML approaches has yet to be addressed. This paper has failed to find compelling works which describe how the data visualization capabilities of SOMs can be leveraged to detect and analyze hidden patterns in illicit transactions. Future research will aim to include how SOMs can be used in relation with AML systems.

V. CONCLUSION

The goal of this paper was to review the efforts of researchers utilizing machine learning approaches to combat money laundering. Herein, traditional AML implementation approaches were briefly discussed, as well as their limitations. Recently, machine learning technologies have been used to further streamline the AML process as a solution to the aforementioned method. According to the researchers' work, some existing ML approaches have been reviewed and compared in terms of their merits and demerits. In order to combat, monitor, and analyze illicit financial activity, financial institutions use ever-evolving ML/DL-based AML and risk management systems. Experts are combining semantic web, knowledge graph (KG), and graph neural networks (GNN) technologies with transfer learning ML/DL to deliver more optimal results and faster computation with very huge datasets in order to improve the performance of AML and risk analysis systems.

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